Optimal Execution

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Optimal Execution

A frequently encountered problem in finance is to buy (acquire) or sell (liquidate) a large amount of shares.

- Example: an agency broker who trade on behalf of institutional investors, e.g., pension funds, hedge funds, sovereign funds, etc.

The trader typically slices the large order into smaller parts, facing a trade-off:

- Trading faster: larger price impact.
- Trading slower: higher price risk

Execution cost: the difference between the benchmark price and actual price.

- Known as the implementation shortfall or slippage if the benchmark price is taken as the midprice.

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The Model

The basic model of using MOs to perform liquidation or acquisition includes the following:

- $\nu = (\nu_t)_{0\le t\le T}$: the trading rate/speed.
- $Q^\nu = (Q^\nu_t)_{0\le t\le T}$: the agent's inventory.
- $S^\nu = (S^\nu_t)_{0\le t\le T}$: the midprice.
- $\hat{S}^\nu = (\hat{S}^\nu_t)_{0\le t\le T}$: the execution price.
- $X^\nu = (X^\nu_t)_{0\le t\le T}$: the agent's cash amount.

The inventory process satisfies that:

$$dQ^\nu_t = \pm\nu_t dt, \qquad Q^\nu_0 = q.$$

The Permanent and Temporary Price Impact

The midprice/fundamental/efficient process:

$$dS_t^\nu = \pm g(\nu_t)dt + \sigma dW_t, \qquad S_0^\nu = S,$$

where $(W_t)_{0\le t\le T}$ is a standard Brownian motion and $g : \mathbb{R}_+ \to \mathbb{R}_+$ is the permanent price impact.

The execution price process:

$$\hat{S}_t^\nu = S_t^\nu \pm (\Delta/2 + f(\nu_t)), \qquad \hat{S}_0^\nu = \hat{S},$$

where $\Delta \ge 0$ is the bid-ask spread, and $f : \mathbb{R}_+ \to \mathbb{R}_+$ is the temporary price impact.

The Cash Process and Expected Revenue

The cash process:

$$dX_t^\nu = \hat{S}_t^\nu \nu_t dt, \qquad X_0^\nu = x.$$

The expected revenue:

$$R^\nu = \mathbb{E}[X_T^\nu] = \mathbb{E}\left[\int_0^T \hat{S}_t^\nu \nu_t dt\right].$$

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Liquidation with Only Temporary Price Impact

We assume that $\Delta = g(\nu) = 0$.

The execution cost:

$$EC^{\nu} = Q_0 S_0 - \mathbb{E}\left[\int_0^T \hat{S}_t^{\nu} \nu_t dt\right].$$

Minimizing the execution cost is equivalent to maximizing the execution revenue $\mathbb{E}[\int_0^T \hat{S}_t^{\nu} \nu_t dt]$.

The agent's value function:

$$H(t, S, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}\left[\int_t^T \hat{S}_u^{\nu} \nu_u du\right] = \sup_{\nu \in \mathcal{A}} \mathbb{E}\left[\int_t^T (S_u - k\nu_u)\nu_u du\right].$$

The HJB Equation and Boundary Conditions

It is straightforward to write down the HJB equation:

$$H_t + \frac{1}{2}\sigma^2 H_{SS} + \sup_{\nu}\left\{(S - k\nu)\nu - \nu H_q\right\} = 0.$$

Boundary conditions:

- Liquidating all the inventory by T:

$$H(T, S, q) \to -\infty, \qquad q > 0.$$

- No need to trade with no inventory:

$$H(t, S, 0) = 0.$$

Solving the HJB Equation

Applying the FO to the HJB equation, we get

$$\nu^* = \frac{1}{2k}(S - H_q).$$

The HJB equation reduces to

$$H_t + \frac{1}{2}\sigma^2 H_{SS} + \frac{1}{4k}(S - H_q)^2 = 0.$$

We propose the ansatz: $H(t, S, q) = qS + h(t, q)$. Then

$$h_t + \frac{1}{4k}(h_q)^2 = 0,$$

with $h(T, q) = -\infty$ for $q > 0$ and $h(t, 0) = 0$.

Solving the HJB Equation (continued)

We propose another ansatz: $h(t,q) = q^2 h_2(t)$. Then

$$h_2'(t) + \frac{1}{k}\big(h_2(t)\big)^2 = 0,\ \ h_2(T) = -\infty.$$

The solution to the ODE above:

$$h_2(t) = \left(\frac{1}{h_2(T)} - \frac{1}{k}(T-t)\right)^{-1} = \frac{k}{t-T}.$$

Then

$$\nu_t^* = -\frac{1}{k} h_2(t) Q_t^{\nu^*}.$$

Using that $dQ_t^{\nu^*} = -\nu_t^* dt$ and we have

$$\int_0^t \frac{dQ_s^{\nu^*}}{Q_s^{\nu^*}} = \int_0^t \frac{h_2(s)}{k} ds \Rightarrow Q_t^{\nu^*} = Q_0 \frac{T-t}{T},\ \ \nu_t^* = Q_0/T,$$

which coincides with the time-weighted average price (TWAP).

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Optimal Acquisition with Terminal Penalty and Temporary Price Impact

Assumptions: the trader is to acquire $\mathcal{R}$ shares of an asset, $f(\nu) = k\nu$, $g(\nu) = \Delta = 0$.

Expected cost:

$$\mathbb{E}\left[\int_0^T \hat{S}_u^\nu \nu_u du + (\mathcal{R} - Q_T^\nu) S_T + \alpha(\mathcal{R} - Q_T^\nu)^2\right],$$

where the the last term is the terminal penalty.

Let $Y_t^\nu = \mathcal{R} - Q_t^\nu$ ($dY_t = -\nu_t dt$). The value function can be written as

$$H(t, S, y) = \inf_{\nu \in \mathcal{A}} \mathbb{E}_{t,S,y}\left[\int_t^T \hat{S}_u^\nu \nu_u du + Y_T^\nu S_T + \alpha(Y_T^\nu)^2\right].$$

The HJB Equation

The HJB equation:

$$H_t + \frac{1}{2}\sigma^2 H_{SS} + \inf_{\nu}\left\{(S + k\nu)\nu - \nu H_y\right\},$$

with the terminal condition $H(T, S, y) = yS + \alpha y^2$.

The FO implies that

$$\nu^* = \frac{1}{2k}(H_y - S).$$

The HJB equation reduces to that

$$H_t + \frac{1}{2}\sigma^2 H_{SS} - \frac{1}{4k}(H_y - S)^2 = 0.$$

Solving the HJB Equation

We apply the ansatz:

$$H(t,S,y) = yS + h_0(t) + h_1(t)y + h_2(t)y^2,$$

with $h_0(T) = h_1(T) = 0$ and $h_2(T) = \alpha$.

Then,

$$\begin{aligned}&\big(h_2'(t) - h_2(t)^2/k\big)y^2 + \big(h_1'(t) - h_1(t)h_2(t)/(2k)\big)y \\ &+ \big(h_0'(t) - h_1(t)^2/(4k)\big) = 0.\end{aligned}$$

Then $h_0(t) = h_1(t) = 0$, and

$$h_2(t) = \big((T-t)/k + 1/\alpha\big)^{-1}.$$

The solution converges to TWAP when $\alpha \to 0$.

The Optimal Strategy

Then

$$\nu_t^* = \left((T-t)/k + 1/\alpha\right)^{-1} Y_t^{\nu^*}.$$

Therefore $dY_t^{\nu^*} = -\nu_t^* dt$ implies that

$$Q_t^* = \mathcal{R} - Y_t^* = \frac{t}{T + k/\alpha}\mathcal{R}.$$

Moreover, the optimal acquisition strategy is as follows:

$$\nu_t^* = \frac{\mathcal{R}}{T + k/\alpha}.$$

$\alpha \to 0$: Only acquisite at T.

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Optimal Liquidation with Unbalanced Order Flow

Let the buy/sell order flows be $\mu_t^{\pm}$ and we assume that they satisfy the following SDE:

$$d\mu_t^{\pm} = -\kappa\mu_t^{\pm}dt + \eta dL_t^{\pm}, \qquad \mu_0^{\pm} = \mu^{\pm},$$

where $L_t^{\pm}$ are independent Poisson processes with equal intensity λ.

The net order flow:

$$d\mu_t = d\mu_t^{+} - d\mu_t^{-} = -\kappa\mu_t dt + \eta(dL_t^{+} - dL_t^{-}).$$

The net order flow process is driven by a compound Poisson process.

The Midprice Process

The midprice process:

$$dS_t^\nu = \big(g(\mu_t^+) - g(\mu_t^- + \nu_t)\big)\,dt + \sigma\,dW_t, \qquad S_0^\nu = S.$$

Suppose $g(\nu) = b\nu$, then

$$\begin{aligned} dS_t^\nu &= \big(b\mu_t^+ - b\mu_t^- - b\nu_t\big)\,dt + \sigma\,dW_t \\ &= \big(b(\mu_t^+ - \mu_t^-) - b\nu_t\big)\,dt + \sigma\,dW_t \\ &= \big(b\mu_t - b\nu_t\big)\,dt + \sigma\,dW_t. \end{aligned}$$

The inventory process:

$$dQ_t^\nu = -\nu_t dt.$$

The cash value process:

$$dX_t^\nu = \hat{S}_t^\nu \nu_t dt = (S_t^\nu - k\nu_t)\nu_t dt.$$

The Value Function and HJB Equation

The value function:

$$H(t,x,S,\mu,q)=\sup_{\nu\in\mathcal{A}}\mathbb{E}_{t,x,S,\mu,q}\left[X_T^\nu+Q_T^\nu(S_T-\alpha Q_T^\nu)-\phi\int_t^T(Q_s^\nu)^2ds\right].$$

The HJB equation:

$$\begin{aligned}&H_t+\frac{1}{2}\sigma^2H_{SS}+\mathcal{L}^\mu H-\phi q^2\\&+\sup_\nu\left\{(S-k\nu)\nu H_x+b(\mu-\nu)H_S-\nu H_q\right\}=0,\end{aligned}$$

with the terminal condition $H(T,x,S,\mu,q)=x+qS-\alpha q^2$, where

$$\begin{aligned}\mathcal{L}^\mu H=&-\kappa\mu H_\mu+\lambda\big[H(t,x,S,\mu+\eta,q)-H(t,x,S,\mu,q)\big]\\&+\lambda\big[H(t,x,S,\mu-\eta,q)-H(t,x,S,\mu,q)\big].\end{aligned}$$

Solving the HJB Equation

We apply the ansatz:

$$H(t,x,S,\mu,q) = x + qS + h(t,\mu,q).$$

Then,

$$h_t + \mathcal{L}^{\mu} h + b\mu q - \phi q^2 + \sup_{\nu} \left\{ -k\nu^2 - (bq + h_q)\nu \right\} = 0,$$

with the terminal condition $h(T,\mu,q) = -\alpha q^2$.

The FO implies that

$$\nu^* = -\frac{1}{2k}(bq + h_q).$$

Moreover,

$$h_t + \mathcal{L}^{\mu} h + b\mu q - \phi q^2 + \frac{(bq + h_q)^2}{4k} = 0.$$

Solving the HJB Equation (continued)

We apply the ansatz:

$$h(t,\mu,q) = h_0(t,\mu) + h_1(t,\mu)q + h_2(t,\mu)q^2.$$

Then,

$$(\partial_t + \mathcal{L}^\mu)h_0 + \frac{1}{4k}h_1^2 = 0,$$
$$(\partial_t + \mathcal{L}^\mu)h_1 + b\mu + \frac{1}{2k}h_1(b+2h_2) = 0,$$
$$(\partial_t + \mathcal{L}^\mu)h_2 - \phi + \frac{1}{4k}(b+2h_2)^2 = 0,$$

with the terminal conditions $h_0(T,\mu) = h_1(T,\mu) = 0$ and $h_2(T,\mu) = -\alpha$.

Solving the HJB Equation (continued)

We first note that h_2 is indepnedent of μ. Then the solution to the ODE for h_2 is:

$$h_2(t,\mu) = \chi(t) - b/2,\ \chi(t) = \sqrt{k\phi}\frac{1+\zeta e^{2\gamma(T-t)}}{1-\zeta e^{2\gamma(T-t)}},$$

where $\gamma = \sqrt{\phi/k}$ and $\zeta = \frac{\alpha - b/2 + \sqrt{k\phi}}{\alpha - b/2 - \sqrt{k\phi}}$.
For h_1, we apply the ansatz:

$$h_1(t,\mu) = \ell_0(t) + \ell_1(t)\mu.$$

Then,

$$\left(\ell_0'(t) + \chi(t)\ell_0(t)/k\right) + \Big(\ell_1'(t) + b + \big(\chi(t)/k - \kappa\big)\ell_1(t)\Big)\mu = 0,$$

and the terminal condition $\ell_0(T) = \ell_1(T) = 0$.

The Optimal Strategy

Solving the ODEs for ℓ_0 and ℓ_1, we get:

$$\ell_0(t) = 0, \ell_1(t) = b\bar{\ell}_1(\tau),$$

where

$$\bar{\ell}_1(\tau) = \frac{1}{\zeta e^{\gamma\tau} - e^{-\gamma\tau}} \left\{ e^{\gamma\tau} \frac{1 - e^{-(\kappa+\gamma)\tau}}{\kappa + \gamma} - e^{-\gamma\tau} \frac{1 - e^{-(\kappa-\gamma)\tau}}{\kappa - \gamma} \right\}$$

The optimal strategy is as follows:

$$\nu_t^* = -\frac{1}{k}\chi(t) Q_t^{\nu^*} - \frac{b}{2k}\bar{\ell}_1(t)\mu_t,$$

which sometimes can be negative when μ_t is sufficiently large (to take advantage of the predicitivity of asset price).

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Optimal Liquidation in Lit and Dark Markets

Assumptions:

- The trader is to liquidate $\mathcal{R}$ shares of inventory by T; $g(\nu) = 0$, $f(\nu) = k\nu$, $\Delta = 0$.
- The trader has an access to both the lit and dark markets.
- The midprice process S_t follows a Brownian motion with volatility σ, and trading in the dark market has no price impact.
- Suppose trader posts $y_t \leq Q_t$ units of inventory in the dark market, and y_t is adjusted continuously, and the matching orders in the dark market are independent and identically distributed with volume ξ_j, with arrivals following a Poisson process with intensity λ.

The inventory process:

$$dQ_t^{\nu,y} = -\nu_t dt - \min(y_t, \xi_{1+N_{t-}}) dN_t.$$

The Value Function

The cash value process:

$$dX_t^{\nu,y} = (S_t - k\nu_t)\nu_t dt + S_t \min\left(y_t, \xi_{1+N_{t^-}}\right) dN_t.$$

The value function:

$$H(t,x,S,q) = \sup_{\nu,y\in\mathcal{A}} \mathbb{E}_{t,x,S,q}\left[X_\tau + Q_\tau^{\nu,y}\left(S_\tau - \alpha Q_\tau^{\nu,y}\right) - \phi\int_t^\tau \left(Q_u^{\nu,y}\right)^2 du\right]$$

where the expectation is conditional on $X_{t^-} = x, S_t = S, Q_{t^-} = q$, and the stopping time is defined as

$$\tau = T \wedge \inf\left\{t : Q_t = 0\right\}.$$

The posted volume in the dark pool is at most the remaining inventory, i.e. $y_t \leq Q_t^{\nu,y}$.

The HJB Equation

The HJB equation:

$$\begin{aligned} & H_t + \frac{1}{2}\sigma^2 H_{SS} - \phi q^2 \\ & \quad + \sup_{\nu} \{(S - k\nu)\nu H_x - \nu H_q\} \\ & \quad + \sup_{y \le q} \{\lambda \mathbb{E}[H(t, x + S\min(y,\xi), S, q - \min(y,\xi)) - H]\} = 0, \end{aligned}$$

with the terminal condition $H(T, x, S, q) = x + q(S - \alpha q)$.

We apply the ansatz:

$$H(t, x, S, q) = x + qS + h(t, q).$$

Then, $h(T, q) = -\alpha q^2$ and

$$\begin{aligned} & h_t - \phi q^2 + \sup_{\nu} \left\{-k\nu^2 - \nu h_q\right\} \\ & \quad + \sup_{y \le q} \{\lambda \mathbb{E}[h(t, q - \min(y,\xi)) - h(t, q)]\} = 0. \end{aligned}$$

Solving the HJB Equation

The optimal feedback strategy: $\nu^* = -\frac{1}{2k} h_q(t, q)$.

Then, the HJB equation can be simplied following

$$\sup_{\nu} \left\{ -k\nu^2 - \nu h_q \right\} = \frac{(h_q)^2}{4k}.$$

Solving the HJB Equation (continued)

To derive an analytical solution, we assume that $\xi \geq y_t$ for all t, and hence we only need to consider that $\min(y, \xi) = y$.

We further apply the ansatz: $h(t,q) = h_0(t) + h_1(t)q + h_2(t)q^2$ with $h_0(T) = h_1(T) = 0$ and $h_2(T) = -\alpha$. Then,

$$\begin{aligned}\sup_{y \leq q} \mathbb{E}[h(t, q - \min(y,\xi)) - h(t,q)] &= \sup_{y \leq q}[h(t, q-y) - h(t,q)] \\ &= \sup_{y \leq q}\left[-yh_1 + \left(y^2 - 2qy\right)h_2\right] \\ &= -\frac{1}{4h_2}\left(h_1 - 2qh_2\right)^2,\end{aligned}$$

The optimal strategies are given by:

$$y^* = q + \frac{h_1}{2h_2},$$

$$\nu^* = -\frac{1}{2k}(h_1 + 2qh_2).$$

Solving the HJB Equation (continued)

Substituting the ansatz back to the HJB equation, we get:

$$\partial_t h_2 - \phi - \lambda h_2 + \frac{1}{k} h_2^2 = 0,$$

$$\partial_t h_1 + \left(\lambda + \frac{1}{k} h_2\right) h_1 = 0,$$

$$\partial_t h_0 + \frac{1}{k} h_1^2 - \frac{\lambda}{4} \frac{h_1^2}{h_2} = 0.$$

Let $\zeta^{\pm}$ be the solutions to the quadratic equation $\phi + \lambda p - p^2/k = 0$. Then,

$$h_2'(t) = -\frac{1}{k}(h_2 - \zeta^-)(h_2 - \zeta^+),$$

where $\zeta^{\pm} = k\lambda/2 \pm \sqrt{\lambda^2 k^2/4 + \phi k}$.

This further implies that $\frac{h_2'(t)}{h_2(t)-\zeta^-} - \frac{h_2'(t)}{h_2(t)-\zeta^+} = \frac{1}{k}(\zeta^+ - \zeta^-)$.

Solving the HJB Equation (continued)

Integrating the above equation, we get:

$$h_2(t) = \frac{\zeta^- - \zeta^+ \beta e^{-\gamma(T-t)}}{1 - \beta e^{-\gamma(T-t)}},$$

where $\beta = \frac{\alpha+\zeta^-}{\alpha+\zeta^+}$ and $\gamma = (\zeta^+ - \zeta^-)/k$.
The optimal trading strategy is given by:

$$\nu^* = -h_2(t) Q_t^{\nu^*, y^*}/k, \; y_t^* = Q_t^{\nu^*, y^*}.$$

Then

$$dQ_t^{\nu^*, y^*} = -\nu_t^* dt = -h_2(t) Q_t^{\nu^*, y^*} dt/k.$$

Integrating the above equation, we get:

$$Q_t^{\nu^*, y^*} = e^{(\zeta^-/k)t} \left(\frac{1 - \beta e^{-\gamma(T-t)}}{1 - \beta e^{-\gamma T}} \right) \mathcal{R}.$$

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Liquidation with Only Limit Orders

Assumptions and notations:

- The trader is to liquidate $\mathcal{R}$ shares of inventory by T, using only limit orders.
- The midprice process S_t follows a Brownian motion with volatility σ: $S_t = S_0 + \sigma W_t$.
- $\delta = (\delta_t)_{0 \le t \le T}$: the depth at which the trader posts limit sell orders.
- $M = (M_t)_{0 \le t \le T}$: the market order flow, a Poisson process with intensity λ denoting the number of market orders arriving by time t.
- $N^\delta = (N^\delta_t)_{0 \le t \le T}$: the number of market buy orders that lift the trader's offer at depth δ_t (i.e., with price greater than or equal to $S_t + \delta_t$).
- $P(\delta) = e^{-\kappa\delta}$: the probability that the LO with depth δ is lifted by a market order.

The Value Function

The cash value process:

$$dX_t = (S_t + \delta_t)dN_t^\delta.$$

Let $\tau = T \wedge \inf\{t : Q_t^\delta = 0\}$. Then, we define the value function as:

$$H(t, x, S, q) = \sup_{\delta \in \mathcal{A}} \mathbb{E}_{t,x,S,q}[X_\tau^\delta + Q_\tau^\delta(S_\tau - \alpha Q_\tau^\delta)],$$

where the expectation is conditional on $X_{t^-} = x, S_t = S, Q_{t^-} = q$.

The HJB Equation

The HJB equation:

$$\partial_t H + \frac{1}{2}\sigma^2 \partial_{SS} H + \sup_{\delta} \left\{ \lambda e^{-\kappa\delta} [H(t, x + (S+\delta), S, q-1) - H(t,x,S,q)] \right\} = 0,$$

$$H(t,x,S,0) = x,$$

$$H(T,x,S,q) = x + q(S - \alpha q).$$

Solving the HJB Equation

We apply the ansatz: $H(t,x,S,q) = x + qS + h(t,q)$. Then,

$$h_t + \sup_{\delta} \left\{ \lambda e^{-\kappa\delta} [\delta + h(t,q-1) - h(t,q)] \right\} = 0,$$

with the boundary conditions $h(T,q) = -\alpha q^2$ and $h(t,0) = 0$.

By the FO, we have that

$$\delta^* = \frac{1}{\kappa} + h(t,q) - h(t,q-1).$$

Solving the HJB Equation (continued)

Subustituting the optimal strategy back to the HJB equation, we get:

$$h_t + \frac{\tilde{\lambda}}{\kappa} exp\Big\{ -\kappa\big[h(t,q) - h(t,q-1)\big]\Big\} = 0,$$

where $\tilde{\lambda} = \lambda/e$.

Let $h(t,q) = \ln(\omega(t,q))/\kappa$. Then

$$\omega_t(t,q) + \tilde{\lambda}\omega(t,q-1) = 0,$$

with the boundary condition $\omega(t,0) = 1$ and $\omega(T,q) = e^{-\kappa\alpha q^2}$.

Solving the HJB Equation (continued)

The solution to the ODE above is:

$$\omega(t,q) = \sum_{n=0}^{q} \frac{\tilde{\lambda}^n}{n!} e^{-\kappa\alpha(q-n)^2} (T-t)^n.$$

The optimal depth is given by:

$$\delta^* = \frac{1}{\kappa}\left(1 + \ln\left(\frac{\omega(t,q)}{\omega(t,q-1)}\right)\right).$$